

ECR-2 Counter-Current Simulation Report

Frozen calculation snapshot for preliminary engineering review

RUN #845

WARNING / PRELIMINARY ENGINEERING

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Three accepted preliminary D/H trials; two smaller trials dependency-blocked.

Executive result

Run #845 used the current Stage 4 Process Design basis: 50 deg C extraction temperature, RRBO composition 85/5/5/5 wt%, RRBO feed 3,448 kg/h, NMP feed 4,024 kg/h, and a 10 mol% hydrocarbon-only raffinate aromatic target.

Interpretation

The run is warning-status because $D = 0.30$ m and $D = 0.45$ m fail the K&H 1995 holdup physical-validity check. $D = 0.60$ m, 0.80 m, and 1.00 m each converged and passed mass balance.

Best preliminary trial (not a final selection)

Diameter	0.60 m
Required active height	0.279375 m
RRBO recovery	90.3339%
Raffinate aromatics	9.9989% hydrocarbon-only mol fraction
BVP / mass balance	converged / passed

ECR-2 generates the complete feasible preliminary (D,H) set. It does not select a final diameter.

1. Frozen inputs and calculation basis

Frozen Run #845 input snapshot

Operating temperature	50 deg C (Stage 4 Extraction Temperature)
RRBO feed	3448.000 kg/h
NMP feed	4024.000 kg/h
S/O ratio	1.167053 mass basis
Phase configuration	NMP continuous / RRBO dispersed
Theoretical stages, N_T	1 (calculated)
Raffinate target	10 mol% aromatics, hydrocarbon-only physical outlet basis

RRBO feed composition

Component	Mass fraction	Feed flow
Saturates	85 wt%	2930.800 kg/h
Mono-aromatics	5 wt%	172.400 kg/h
Di-aromatics	5 wt%	172.400 kg/h
Poly-aromatics	5 wt%	172.400 kg/h
Total aromatics	15 wt%	517.200 kg/h
RRBO total	100 wt%	3448.000 kg/h

ECR-2 generated sizing basis

Specific agitation psi	0.1000 W/kg
d32 route	DIRECT_TURBULENCE_D32_PRELIMINARY
d32 result	0.976126 mm (C = 0.39)
C sensitivity	0.901 to 1.076 mm d32
Diameter trials	0.30, 0.45, 0.60, 0.80, 1.00 m
Height method	Progressive full-BVP trials with conservative upper bracket endpoint
Numerical mesh	Maximum delta-z = 0.25 m; physical height is independent of mesh size

Equation basis: $d_{32} = C * (\gamma / \rho_c)^{0.6} * \epsilon^{(-0.4)}$, with $\epsilon = \text{psi} = 0.1 \text{ W/kg}$. The direct-turbulence d32 is preliminary, not pilot validated, and separate from K&H 1996.

2. Governing equations and calculation chain

Equations used for Run #845

The equations below are the implemented calculation structures for this frozen snapshot. Where a correlation or evidence gate is preliminary or blocked, the equation is shown for traceability and is not presented as governed design data.

1. Physical feed composition

$$m_{i,RRBO} = w_{i,RRBO} * m_{RRBO}; \quad m_{RRBO} = 3448 \text{ kg/h}$$

For Run #845: $w = [0.85, 0.05, 0.05, 0.05]$ for [Sat, Mono, Di, Poly].

2. Solvent-to-oil ratio

$$S/O_{\text{mass}} = m_{\text{NMP}} / m_{\text{RRBO}} = 4024 / 3448 = 1.167053$$

The report uses the mass-basis ratio; no legacy manual rotor-power or diameter input is used.

3. Physical mass concentration conversion

$$w_i = z_i * MW_i / \sum(z_j * MW_j); \quad C_i = w_i * \rho_{\text{phase}}$$

Physical molecular weights convert NRTL mole fractions to transport concentrations; they do not modify NRTL coordinates.

4. K&H 1995 characteristic scale

$$\theta = (\rho_c / (g * \sigma))^{0.25}$$

θ has units s/m; $g = 9.80665 \text{ m/s}^2$ and σ is interfacial tension.

5. K&H 1995 dispersed holdup

$$\phi_{\text{raw}} = [0.0267 + (\psi * \theta / g)^{0.77}] * (U_d * \theta)^{0.64} * \exp(20.7 * U_c * \theta)^{0.90} * ((\rho_c - \rho_d) / \rho_c)^{(-0.34)} * 2.27 * x_f^{(-0.77)}$$

Physical guard: only $0 < \phi_{\text{raw}} < 1$ is usable. Run #845 records $\phi_{\text{raw}} = 10.4189$ for blocked 0.30/0.45 m initial trials; no clamping is applied.

6. Direct-turbulence d32

$$d_{32} = C * (\sigma / \rho_c)^{0.6} * \epsilon^{(-0.4)}; \quad \epsilon = \psi = 0.1 \text{ W/kg}$$

$C = 0.39$; calculated $d_{32} = 0.000976126 \text{ m} = 0.976126 \text{ mm}$. Preliminary only; C sensitivity is 0.36 to 0.43.

7. Specific interfacial area

$$a = 6 * \phi_d / d_{32}$$

a is in m^2/m^3 . It is available only when both usable holdup and usable d_{32} are present.

2. Governing equations - transfer and BVP

Local K&H transfer equations

8. Dimensionless groups

```
Re_d = rho_c * U_slip * d32 / mu_c;   kappa = mu_d / mu_c
Sc_c = mu_c / (rho_c * De_c);   Sc_d = mu_d / (rho_d * De_d);   Pe_c = d32 * U_slip / De_c
```

These are local, component-aware quantities. Diffusivities and their provenance are required inputs.

9. Continuous-phase Sherwood number

```
Sh_c,r = 2.43 + 0.775*Re_d^0.5*Sc_c^(1/3) + 0.0103*Re_d*Sc_c^(1/3)
Sh_c,infinity = 50 + (2/sqrt(pi))*Pe_c^0.5
R = 0.0526*Re_d^(1/3+0.0659*Re_d^0.25)*Sc_c^(1/3)*(U_slip*mu_c/sigma)^(1/3)
    * [1/(1+kappa^1.1)] * [1 + C1*((psi/g)*(rho_c/(g*sigma)))^0.25]^(1/3)]
Sh_c = (1-phi_d)*(Sh_c,r + R*Sh_c,infinity)/(1+R)
```

Preliminary K&H 1999 local-kernel structure; scoped Kuhni C1 = 7.5.

10. Dispersed-phase Sherwood number

```
q = Re_d * Sc_d^(1/3)
Sh_d = 17.7 + [0.00319*q^1.7/(1+0.0143*q^0.7)] * (rho_d/rho_c)^(2/3) / (1+kappa^(2/3))
```

Preliminary K&H 1999 local-kernel structure.

11. Film and overall coefficients

```
k_c = Sh_c*De_c/d32;   k_d = Sh_d*De_d/d32
K_d = C_d,i* / C_c,i*;   K_overall = k_c*k_d/(K_d*k_d + k_c)
K_oa = K_overall*a
```

K_d is a concentration-basis partition coefficient; it is not substituted by a mole-fraction ratio.

12. Local driving force and transfer rate

```
DeltaC_d = C_d - K_d*C_c;   N_i = K_oa*DeltaC_d
```

Positive transfer direction is RRBO/dispersed to NMP/continuous. These outputs remain subject to the governing evidence gates.

13. Compartment active volume

```
A_column = pi*D^2/4;   Delta z = H/N;   V_j = A_column*Delta z
T_i,j = lambda * N_i,j * V_j * 3600
```

T_i,j is the component transfer amount in kg/h; 3600 converts seconds to hours.

14. Counter-current BVP residuals

```
D_in,i,j - D_out,i,j - T_i,j = 0
C_in,i,j - C_out,i,j + T_i,j = 0
```

The nonlinear solver adjusts the 10N internal face-flow variables; local transfer, properties, holdup, and equilibrium are recalculated at every trial state.

15. Global mass-balance acceptance

```
B_i = (RRBO_feed_i + NMP_feed_i) - D_out,i - C_out,i
B_total = SUM(B_i);   pass if component and total tolerances are satisfied
```

Run #845 accepted trials report passed total balance and converged solver status.

16. Product quality, recovery, and height selection

```
x_aron,raff = SUM(n_aron,raff) / SUM(n_hydrocarbon,raff)
RRBO recovery = SUM(m_Sat..m_Poly,raff) / m_RRBO,feed * 100
H_required = conservative upper endpoint of the refined bracket where x_aron,raff <= 0.10
```

NMP is excluded from both product-quality denominators. Each height trial is a full BVP solve; the selected physical height is independent of numerical cell count.

3. Diameter and height search

Generated diameter trials

Each diameter is evaluated independently. Every physical-height trial reruns the full counter-current BVP; no local NRTL, holdup, d32, or transfer value is reused between height trials.

Diameter	Trial status	Required H	Product aromatics	Cells	Iter.	F evals	Total balance
0.30 m	DEPENDENCY BLOCKED	NOT CALCULATED	—	—	0	0	—
0.45 m	DEPENDENCY BLOCKED	NOT CALCULATED	—	—	0	0	—
0.60 m	FEASIBLE PRELIMINARY	0.279375 m	9.9989%	2	11	232	2.5074e-9 kg/h
0.80 m	FEASIBLE PRELIMINARY	0.385625 m	9.9984%	2	11	232	2.4026e-9 kg/h
1.00 m	FEASIBLE PRELIMINARY	0.412500 m	9.9999%	2	11	232	1.3630e-8 kg/h

Height-sizing method

For accepted trials, the target bracket is refined until the product-quality condition is met. The conservative upper bracket endpoint is selected as the required active extraction height. The selected heights below are physical equipment heights, not numerical cell counts or delta-z values.

D (m)	Selected H (m)	Bracket / selection note
0.60	0.279375	Target bracket refined to [0.2788, 0.2794] m; conservative upper bound selected.
0.80	0.385625	Target bracket refined to [0.3850, 0.3856] m; conservative upper bound selected.
1.00	0.412500	Target bracket refined to [0.4119, 0.4125] m; conservative upper bound selected.

4. Accepted BVP calculations

Accepted preliminary trial diagnostics

The following BVP results satisfy convergence and total mass-balance acceptance for the individual diameter trial. They remain preliminary and are not release-eligible.

D = 0.60 m | H = 0.279375 m

Product aromatics	9.9989% (target 10 mol%)
BVP status	converged; mass balance passed
Solver diagnostics	11 iterations; 232 function evaluations; final residual norm 1.3766e-12
Total mass balance	2.5074e-9 kg/h
RRBO recovery	90.3339%

D = 0.80 m | H = 0.385625 m

Product aromatics	9.9984% (target 10 mol%)
BVP status	converged; mass balance passed
Solver diagnostics	11 iterations; 232 function evaluations; final residual norm 1.0331e-12
Total mass balance	2.4026e-9 kg/h
RRBO recovery	89.7098%

D = 1.00 m | H = 0.412500 m

Product aromatics	9.9999% (target 10 mol%)
BVP status	converged; mass balance passed
Solver diagnostics	11 iterations; 232 function evaluations; final residual norm 4.1178e-12
Total mass balance	1.3630e-8 kg/h
RRBO recovery	89.5699%

RRBO recovery calculation

RRBO recovery = (Saturates + Mono-aromatics + Di-aromatics + Poly-aromatics in raffinate) / RRBO feed x 100. NMP is excluded from RRBO recovery.

Diameter	RRBO raffinate	RRBO transferred to extract	RRBO recovery
0.60 m	3114.712 kg/h	333.288 kg/h	90.3339%
0.80 m	3093.196 kg/h	354.804 kg/h	89.7098%
1.00 m	3088.370 kg/h	359.630 kg/h	89.5699%

5. Outlet composition and component calculations

Accepted raffinate outlet component flows

Component order is Saturates / Mono-aromatics / Di-aromatics / Poly-aromatics / NMP. Product-quality aromatics are calculated on the hydrocarbon-only physical raffinate basis; NMP is excluded from both numerator and denominator.

Component	Feed kg/h	Raffinate kg/h	Extract kg/h
Saturates	2930.800	2694.199	236.601
Mono-aromatics	172.400	134.974	37.426
Di-aromatics	172.400	138.984	33.416
Poly-aromatics	172.400	146.555	25.845
NMP	4003.880	31.109	3972.771

D = 0.60 m product quality

9.9989% aromatics (hydrocarbon-only)

Component	Feed kg/h	Raffinate kg/h	Extract kg/h
Saturates	2930.800	2675.535	255.265
Mono-aromatics	172.400	133.799	38.601
Di-aromatics	172.400	138.097	34.303
Poly-aromatics	172.400	145.765	26.635
NMP	4003.880	32.408	3971.472

D = 0.80 m product quality

9.9984% aromatics (hydrocarbon-only)

Component	Feed kg/h	Raffinate kg/h	Extract kg/h
Saturates	2930.800	2671.283	259.517
Mono-aromatics	172.400	133.546	38.854
Di-aromatics	172.400	137.921	34.479
Poly-aromatics	172.400	145.620	26.780
NMP	4003.880	32.849	3971.031

D = 1.00 m product quality

9.9999% aromatics (hydrocarbon-only)

6. Axial transfer diagnostic — D = 0.60 m

Where transfer occurs through the 0.279375 m active height

This diagnostic is transcribed from the accepted D = 0.60 m, H = 0.279375 m frozen BVP state. Each row is a physical axial increment, bottom to top, preserving the frozen Saturates / Mono / Di / Poly / NMP order. Positive transfer is RRBO/dispersed to NMP/continuous. Local quantities are calculated preliminary diagnostics only; governed transfer design values remain unavailable and the run is not release eligible.

Cell	z bottom	z centre	z top	9G	<f@	d32	a
1	0.000000	0.069844	0.139687	0.139687	0.482824	0.976126 mm	2967.798
2	0.139687	0.209531	0.279375	0.139687	0.456098	0.976126 mm	2803.517

Per-increment component transfer

Cell	Component	Driving force	Koa	Increment transfer	Cumulative transfer
1	Saturates	430.82191	0.001715665	105.095183	105.095183
1	Mono-aromatics	66.80151	0.002069206	19.653614	19.653614
1	Di-aromatics	62.77142	0.001990145	17.762296	17.762296
1	Poly-aromatics	51.77753	0.001813449	13.350549	13.350549
1	NMP	-30.33076	0.003917380	-16.893960	-16.893960
2	Saturates	582.41854	0.001588023	131.505660	236.600843
2	Mono-aromatics	65.42913	0.001910397	17.772441	37.426056
2	Di-aromatics	60.02749	0.001834095	15.653966	33.416262
2	Poly-aromatics	52.52989	0.001672797	12.494016	25.844565
2	NMP	-27.60598	0.003621526	-14.215011	-31.108971

Aromatic outlet reconciliation — hydrocarbon-only basis

Basis / quantity	Value	Interpretation
Aromatic feed	517.200000 kg/h (15.000 wt%)	Mono + Di + Poly in RRBO feed
Raffinate aromatics	420.513117 kg/h	Physical raffinate outlet
Extract aromatics	96.686883 kg/h	Physical extract outlet
RRBO feed - raffinate	96.686883 kg/h	Dispersed-side aromatic transfer
Extract - NMP feed	96.686883 kg/h	Continuous-side aromatic transfer
Cell 1 aromatic transfer	50.766459 kg/h	0.000000–0.139688 m
Cell 2 aromatic transfer	45.920424 kg/h	0.139688–0.279375 m
Cumulative axial transfer	96.686883 kg/h	Equals feed minus raffinate and extract minus NMP feed

7. Warnings, acceptance and release status

Run-level status

Overall run status	WARNING
Governed transfer values	UNAVAILABLE
Release status	NOT_RELEASE_ELIGIBLE

Primary dependency blocker

The $D = 0.30$ m and $D = 0.45$ m initial BVPs are blocked because K&H 1995 returns $\phi_{\text{raw}} = 10.4189$, which is physically inadmissible because holdup must be below 1. The simulator fails closed and does not clamp or substitute the value.

Why governed transfer output is unavailable

The accepted local transfer arrays are preliminary solver results only. The direct-turbulence d32 route and local K&H transfer physics are not pilot validated or release eligible. In addition, the primary/global BVP is dependency-blocked by the smaller diameter trial. Therefore the report retains diagnostic data for accepted trials but does not present governed transfer performance as a final design result.

Additional warnings

- NRTL at 50 deg C is outside its calibrated range; equilibrium targets are extrapolated pending validation.
- Interfacial tension uses an unchanged 70 deg C source anchor at the selected 50 deg C condition under a preliminary constant-over-range basis.
- Physical RRBO molecular weights and SN300 EPD tabular values are assumed/provisional.
- NMP purity is retained as a physical plant-boundary input; canonical fresh NMP is used for the C2 thermodynamic coordinate.
- $d_{32} = 0.976$ mm is DIRECT_TURBULENCE_D32_PRELIMINARY and must not be attributed to K&H 1996.
- Sulfur/DBT prediction is not implemented and must not be inferred from aromatic-transfer results.

8. Engineering interpretation

Summary of calculated results

Run #845 is a preliminary forward-model evaluation on the current Stage 4 feed basis. The feasible preliminary design set returned by the generated diameter search is:

Diameter	Required physical height	RRBO recovery	Raffinate aromatics
0.60 m	0.279375 m	90.3339%	9.9989% mol fraction
0.80 m	0.385625 m	89.7098%	9.9984% mol fraction
1.00 m	0.412500 m	89.5699%	9.9999% mol fraction

Use limitation

These results are suitable for preliminary engineering review and comparison of generated diameter/height trials only. The simulator does not select a final diameter, does not establish a release-grade transfer design, and does not provide sulfur or DBT prediction.

The complete frozen source for this report is ECR-2 calculation Run #845, Revision 8. Any change to Stage 4 temperature, feed, composition, solvent ratio, or governed evidence requires a new simulation run and a new report.